



PG-DITISS

Rapid Fire Sheet - Syllabus- Evaluation Guidelines



SUNBEAM

Institute of Information Technology



Placement Initiative

PG-DITISS

Security Concepts

1. Define Cryptography. What are its types and where they are used?
2. Differentiate between IDS and IPS.
3. What is CIA? Explain with examples.
4. What is a Firewall? What are its types?
5. Explain the brute force attack. How to prevent it?
6. Explain CSRF. How to prevent it?
7. What is port scanning? Which tool is used for it?
8. How to reset a password-protected BIOS configuration?
9. What is MITM attack? How to prevent it?
10. What is the main difference between SSL and TLS?
11. What is 2FA? How to implement it for a public website?
12. What is network sniffing? Have you used any tool for sniffing?
13. What is SSH? How to enable passwordless authentication?
14. What is black box testing and white box testing?
15. Explain the concept of cross-site scripting. How it can be prevented?
16. What is Ethical hacking? Which are common tools for Ethical hacking?
17. What is the difference between vulnerability assessment and penetration testing? Which tools you have used for them?
18. What is difference between Malware, Virus, Trojan and Worm?
19. What is Backdoor? What are its types?
20. What is the difference between Threat, Vulnerability, and Risk?

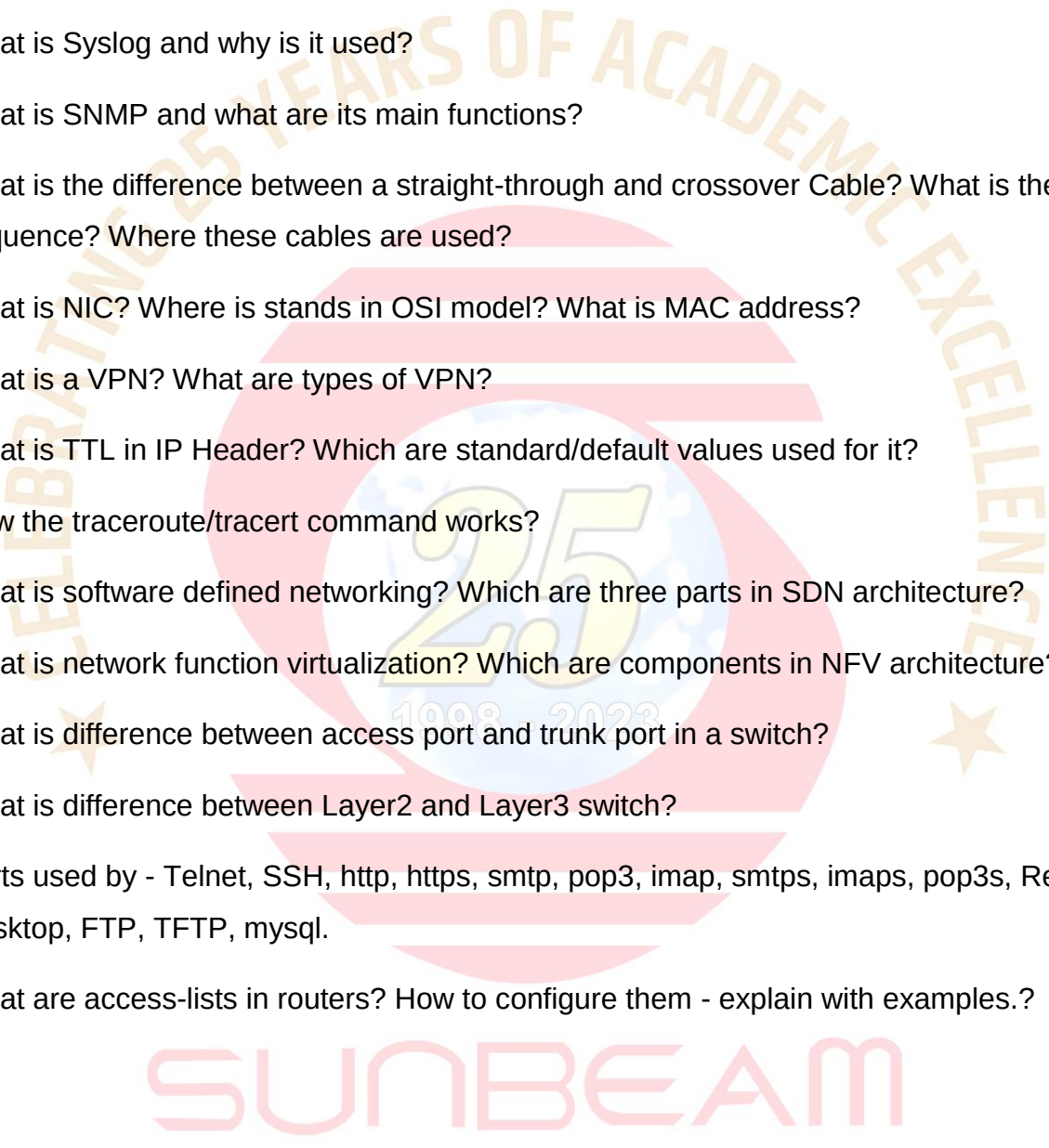
Python Programming

1. What is Python? Name some features of Python and Explain how it different than Java or any other languages?
2. What are the supported data types in python? How to print datatype of a variable?
3. What is difference between list, tuple and dictionary? Is array in C is same as list in Python?
4. Write a program to print square roots of all even numbers in 1 to 100 using list comprehension. Write same code using lambda expressions.
5. What is numpy and how it is better than list in python? What are ways of creating 1D, 2D and 3D arrays in numpy?
6. Write a code to convert List into Tuple in Python and reverse?
7. How can you get list of unique elements from a given list? Write a program.
8. What do you know about Flask & Django? How to implement a simple flask application?
9. How to connect MySQL from Python? Write code for executing DML and DQL queries.
10. What is exception? How to raise and handle exceptions in Python?
11. How to implement abstraction & encapsulation in Python?
12. How to read/write files in Python? Write code using "with" statement.
13. Write a one-liner that will count the number of capital letters in a file.
14. Implement a simple server-client application using Python sockets.
15. Input a directory path from the user. List all text (.txt) files from that directory.

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Fundamentals of Computer Networks

1. What are Private IP address? What are its ranges?
2. Explain the process of three-way handshaking protocol?
3. What is tunneling? Why it is done?
4. What is NAT? Different types of NAT?
5. Which are 7 layers in OSI model? What is the function of the Application Layer in networking?
6. What is the purpose of the Data Link layer? What is the role of the LLC sublayer?
7. What are the different memories used in a CISCO router? What is their purpose? Write command to display RAM contents.
8. What is Ethernet? Explain CSMA/CD.
9. What is difference between BootP and DHCP?
10. What is latency? How can we measure it? What are ways to reduce latencies?
11. What is sub-netting? What is VLSM? If subnet mask is 22, how many hosts can be connected in the network?
12. What is Route Poisoning and split horizon?
13. What are the different types of passwords used in securing a CISCO router? Write commands for different modes.
14. What is difference between IPv4 and IPv6 addressing?
15. State the difference between dynamic IP and static IP addressing?
16. In how many ways you can access the router?
17. What is the difference between TCP and UDP? What are TCP and UDP ports?
18. What are the conversion steps of data encapsulation?
19. What is the administrative distance of RIP, EIGRP and OSPF?
20. What is VTP? What are VTP modes?

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21. What is a VLAN? Why they are created? How to create it? Protocol associated with VLAN?
22. Define STP. Why is it used?
23. What are the major protocols used in WAN? Explain them.
24. What is APIPA? Which address range is used for it?
25. What is Syslog and why is it used?
26. What is SNMP and what are its main functions?
27. What is the difference between a straight-through and crossover Cable? What is their color sequence? Where these cables are used?
28. What is NIC? Where is stands in OSI model? What is MAC address?
29. What is a VPN? What are types of VPN?
30. What is TTL in IP Header? Which are standard/default values used for it?
31. How the traceroute/tracert command works?
32. What is software defined networking? Which are three parts in SDN architecture?
33. What is network function virtualization? Which are components in NFV architecture?
34. What is difference between access port and trunk port in a switch?
35. What is difference between Layer2 and Layer3 switch?
36. Ports used by - Telnet, SSH, http, https, smtp, pop3, imap, smtps, imaps, pop3s, Remote Desktop, FTP, TFTP, mysql.
37. What are access-lists in routers? How to configure them - explain with examples.?

Public Key Infrastructure

1. What are main components of PKI?
2. What is symmetric and asymmetric encryption? What are their advantages and limitations.
3. What is difference between encryption and hashing? Which technique is used in which scenario?
4. Explain public key and private key?
5. What is difference between MD5, SHA and RSA? Why MD5 is not considered as secure?
6. What is X.509? Explain process of attaining digital certificate?
7. What is a certification Authority? What are key functions of CA?
8. Explain hierarchy of CA. What is Intermediate CA and Registration Authority?
9. When your browser gives error about the certificate received from a website? What should be done at website and/or end user to resolve this error?
10. What is Certificate Revocation List (CRL)? Where is Certificate Revocation List stored? How user can view CRL?
11. How the encryption works between your computer and a HTTPS website?
12. What is SSL? Which SSL is secure?
13. What is confusion and diffusion in encryption?
14. Explain the brute force and dictionary attack. How to prevent them?
15. What reciprocal and non-reciprocal cipher? Give examples.
16. What is meaning of x bits' key? Why length of key matters?
17. What is salt? How does it makes password hashing more secure? Any example where you see it used.

Cyber Forensics

1. What is the chain of custody?
2. What steps you will follow at the crime scene while you take control of the computer on the site?
3. How you can identify if a website was accessed from a computer?
4. How to trace the criminal who has created a fake profile of a victim on the Facebook?
5. What precautions you need to take while handling the evidences?
6. How to recover deleted data?
7. What is file shredding? How to do it? How shredding differs from deletion.
8. How to recover data from a physically damaged hard disk?
9. What is file carving?
10. What is MD5 hash collision?
11. What it is required to create hash of all digital evidences?
12. How to recover WhatsApp chats and SMS from mobiles?
13. What do you mean by checksum? What are the popular algorithms for calculating checksum?
What is its significance in cyber forensics?
14. How would you trace a spoofed email?
15. Which tools you have used in cyber forensics?

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Compliance Audit

1. What is a Security Standard? What are guidelines? What are best practices?
2. What is PCI-DSS? What is its purpose? What are the requirements?
3. What is ISMS in ISO27001 security Standard?
4. What is HIPPA? What is its purpose? What are the requirements?
5. What is GDPR? What is its purpose? What are the requirements?
6. What is SOC? Which SOC audit reports are important for service/cloud providers?
7. What are Security audits? Why they are performed?
8. What is a Security policy of an Organization?
9. What is the Risk Assessment? How it is done?
10. What are the advantages and disadvantages of Internal and External Audits?
11. What are CIS controls? How they help organizations?

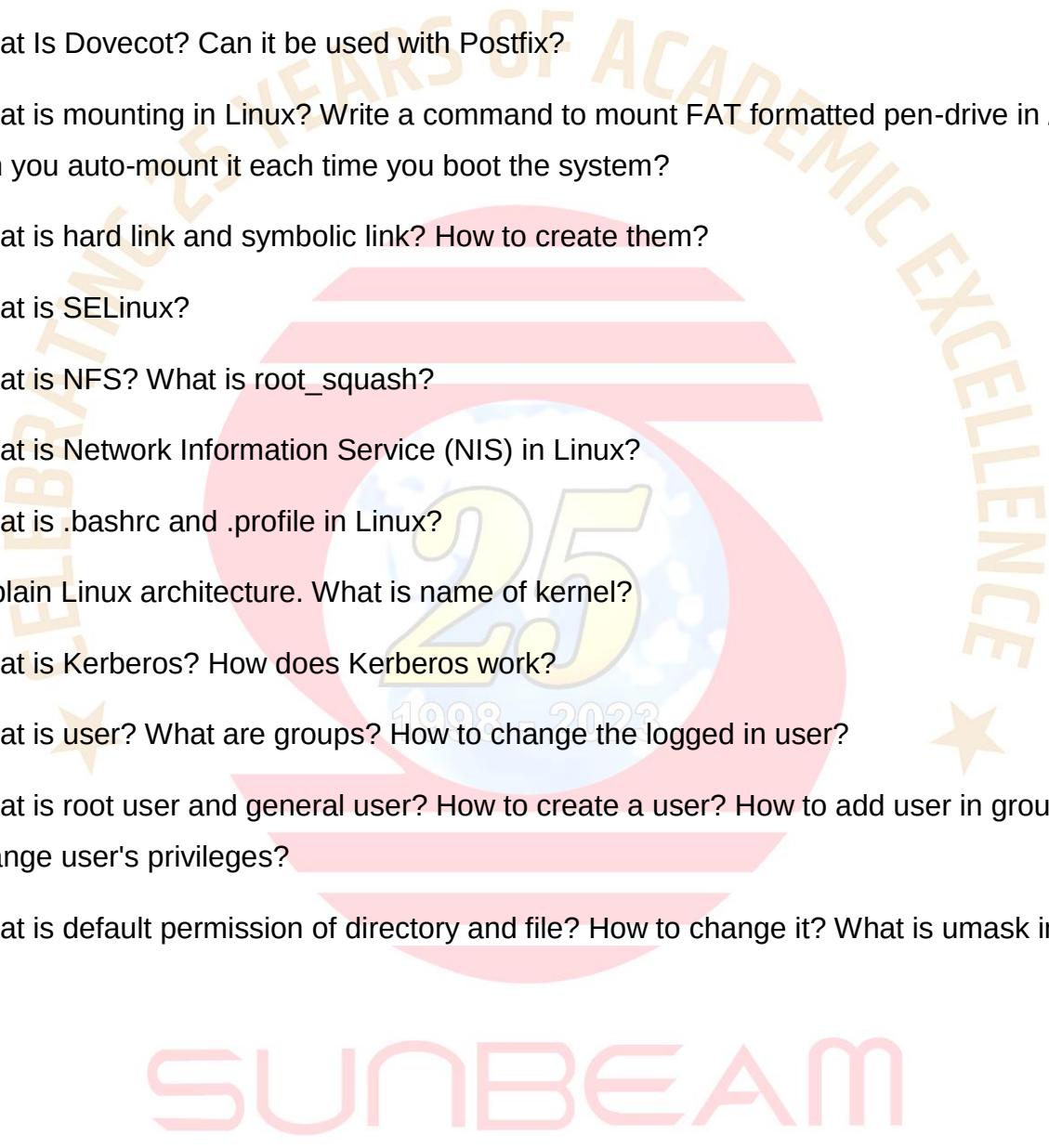
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Network Defense and Countermeasures

1. What is the use of a firewall and how it can be implemented?
2. What is a packet filter firewall?
3. What is a proxy firewall? What are the advantages of using proxy firewall?
4. What is a stateful packet inspection firewall?
5. Why SSL man in the middle is required for proxy firewalls?
6. What are the basic differences between iptables and firewallld? What are the tables used in iptables? Explain in short.
7. What is time based ACL in Squid?
8. What are the types of IDS/IPS? What is false positive and false negative in IDS/IPS?
9. What is snort rule format? Give an example.
10. What is the use of VPN? What is site to site VPN and Dial up VPN?
11. What is IPSec? What is Encapsulating Security Payload in IPSec? What is Authentication Header in IPSec?
12. What is IKE protocol/Framework?
13. What is difference between forward and reverse proxy? What are uses of reverse proxy? Which reverse proxy you have configured?
14. What is phishing and how it can be prevented?
15. How do you defend wireless access point/router at your home?

Concepts of Operating System and Administration

1. Explain Major Functionalities of Operating System.
2. What are different types of CPU Scheduling Algorithms? (FCFS, RR, PRIO, SJF).
3. Explain States of a Process in Operating Systems.
4. Explain Virtual Memory in Operating System. What is Paging/Segmentation?
5. What is difference between process and thread?
6. What is Hyper-V? Explain.
7. What is Active Directory? Explain basic features in Active Directory (AD) of Windows server?
8. What are FSMO roles in Windows Active Directory? How to transfer FSMO roles? What is seize FSMO role option? When to use it?
9. How to take active directory backup? What is authoritative and non-authoritative restore of Active Directory?
10. Explain DNS & DHCP. How to configure in Windows?
11. What is the difference between tree domain vs child domain?
12. What are built in groups - DomainAdmins, Enterprise Admins, Server Operator, Accounts operator, Backup Operator, Domain Users?
13. How to restore a failed Domain controller?
14. Elaborate the concept of Network Policy Server (NPS).
15. What is File Server Resource Manager (FSRM)?
16. What is IIS? Which are the protocols that IIS supports?
17. Explain the concept of Network Load Balancing (NLB).
18. What is PowerShell and what are the features of PowerShell?
19. How to recover forgotten Administrator password in Windows?
20. How to recover forgotten root password in Linux?
21. How to create partition and Filesystem? Write Linux and Windows commands for it.

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22. Explain Linux Directory Structure in detail.
23. What is SSH? What is its default port? Why SSH is secure?
24. Explain each field of /etc/passwd. What is /etc/shadow?
25. Why we use LVM in Linux & what are its benefits?
26. Explain Postfix Mail Server. What is default port used for postfix?
27. What Is Dovecot? Can it be used with Postfix?
28. What is mounting in Linux? Write a command to mount FAT formatted pen-drive in /mnt? How can you auto-mount it each time you boot the system?
29. What is hard link and symbolic link? How to create them?
30. What is SELinux?
31. What is NFS? What is root_squash?
32. What is Network Information Service (NIS) in Linux?
33. What is .bashrc and .profile in Linux?
34. Explain Linux architecture. What is name of kernel?
35. What is Kerberos? How does Kerberos work?
36. What is user? What are groups? How to change the logged in user?
37. What is root user and general user? How to create a user? How to add user in group? How to change user's privileges?
38. What is default permission of directory and file? How to change it? What is umask in Linux??

ITIM and DevOps

1. What is ITIL? What is V3?
2. Explain Software Development Life Cycle.
3. Explore Agile Development. What is scrum?
4. What is Data Center?
5. What is the functionality of a hypervisor?
6. What is Virtualization? What is difference between Type I, Type II virtualization, and containerization?
7. What is cloud computing? Explain cloud service models: IaaS, PaaS, SaaS, FaaS, DaaS.
8. What are the differences between vertical and horizontal scaling? Which one would you prefer and why?
9. How to create EC2 instance in AWS?
10. What is use of these AWS services - EC2, S3, SNS, EKS?
11. What is DevOps? Explain DevOps life cycle.
12. What is GIT? Explain GIT workflow.
13. What is GIT branching? How to implement it? Write commands to create branch, merge branch, and delete branch.
14. What is Docker? Explain Docker architecture.
15. What is Docker image and container? Explain life cycle of Docker container.
16. An application has two components i.e. MySQL database and Python Web application. Explain steps to containerize this application. How to use Docker compose?
17. What is orchestration? Which tools can be used for the same.
18. What is Kubernetes? Explain Kubernetes architecture.
19. What is pod? How do you scale pods?
20. What is CI? What is CD? Which are popular tools for CI/CD pipeline?
21. What is Jenkins? Explain Jenkins architecture.

22. How would you create pipeline in Jenkins?

23. What's the difference between continuous integration, continuous delivery, and continuous deployment?

24. What is Puppet? How it works?

25. What is Nagios? How it works?

26. Explain Service portfolio management.

27. Explain Business relationship management.

28. What is Service Design & its purpose?



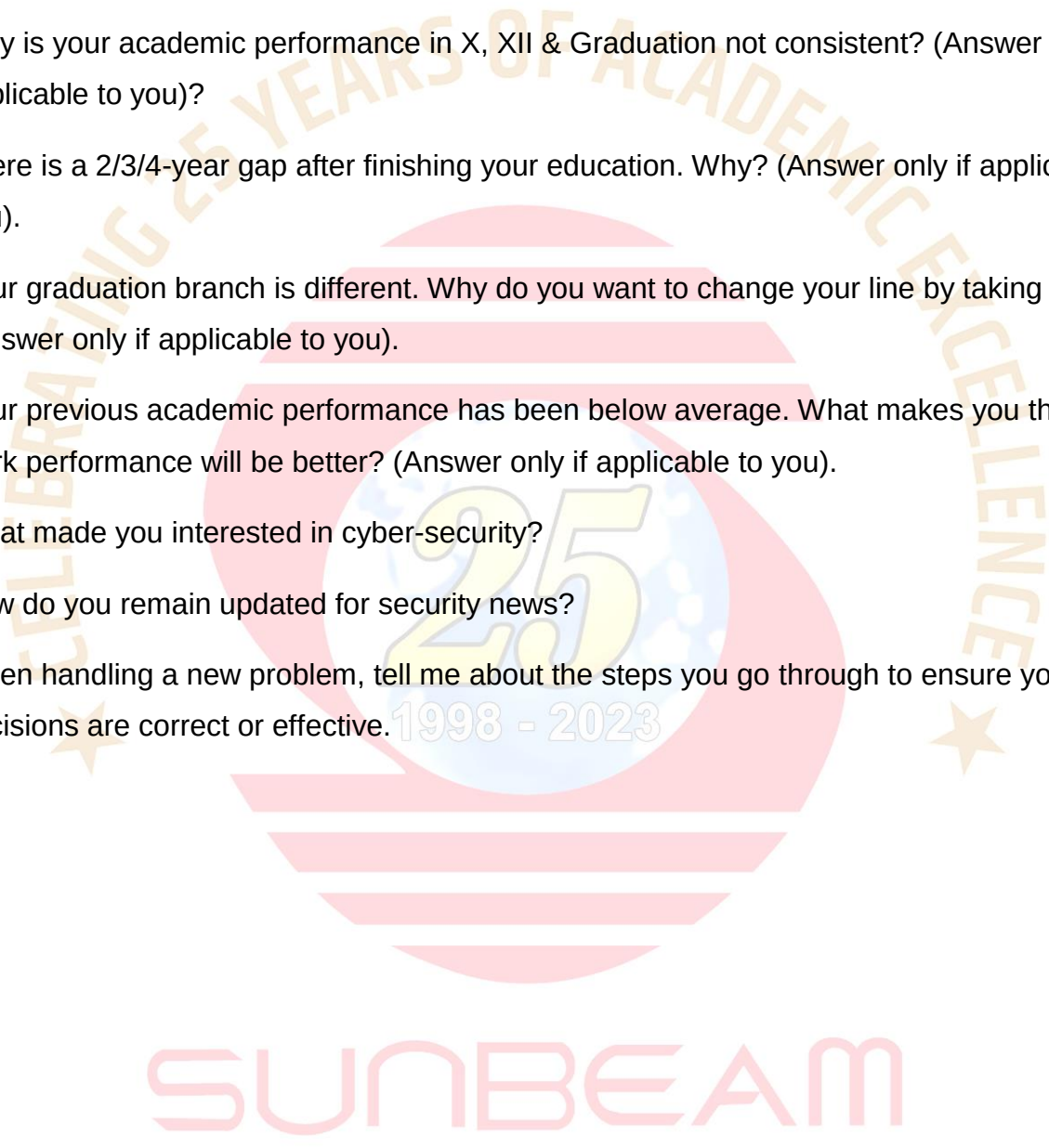
Project Interview Questions

1. Can you explain your C-DAC project? Draw block diagram as appropriate.
2. What's new/different in your project?
3. Explain why did you select this technology & tools for this project?
4. Which advanced features/techniques have you used in your project?
5. What was your role in your project and explain what you did in it?
6. How you have deployed your project in container/cloud?
7. What are the limitations of your project?
8. What are additional functionalities you would like to implement in this project?
9. What are the difficulties you have faced during this project and How you have overcome it?
10. Explain which configuration files you have used?
11. Explain security measures you have implemented in your project?

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HR Interview Questions

1. Tell us something about yourself.
2. Tell us some of your strengths and weaknesses. How you can overcome your weaknesses?
3. Are you comfortable working in a team? What makes you think so?
4. If you were hiring for this position, what qualities would you look for in a potential candidate?
5. How would you rate yourself as a candidate on a scale of 1-10? Give Reasons.
6. What is your career objective?
7. What motivates you at work?
8. What is your opinion about working in night shifts or over the weekends?
9. Do you think you have made the right career choice? Why?
10. Discuss the most stressful situation you came across in life.
11. Why should we hire you?
12. What is your idea of an ideal company?
13. Tell us something about your achievements during your education.
14. Do you think your marks in academics truly reflect your potential?
15. Tell us something about your hobbies.
16. How do you see yourself 5 years from now?
17. Tell us about your dream job.
18. Is there any particular kind of person you cannot work with?
19. Tell us what do you know about our company?
20. How long do you think you can work for our company?
21. Are you willing to sign a service bond with our company? If yes for what duration? If no, why not?

22. How important would the location of your job be for you? Why?
23. Which is the subject that you have liked in your studies till date? Why?
24. What are the things that might make you leave a job?
25. We are hiring for test engineer position. Are you willing to join? Why?
26. What is the reason for the gap in your education? (Answer only if applicable to you)
27. Why is your academic performance in X, XII & Graduation not consistent? (Answer only if applicable to you)?
28. There is a 2/3/4-year gap after finishing your education. Why? (Answer only if applicable to you).
29. Your graduation branch is different. Why do you want to change your line by taking this job? (Answer only if applicable to you).
30. Your previous academic performance has been below average. What makes you think your work performance will be better? (Answer only if applicable to you).
31. What made you interested in cyber-security?
32. How do you remain updated for security news?
33. When handling a new problem, tell me about the steps you go through to ensure your decisions are correct or effective.
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- A large, semi-transparent watermark logo is centered on the page. It features a circular emblem with a globe in the center, surrounded by horizontal lines. Above the globe, the text 'CELEBRATING 25 YEARS OF ACADEMIC EXCELLENCE' is written in a circular path. Below the globe, the year '1998 - 2023' is displayed. At the bottom of the emblem, the word 'SUNBEAM' is written in a stylized, blocky font.